

Dhilipan Sivasamy

ENGINEER



CONTACT ME



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PROFILE

Passionate and results-driven Mechanical Engineer with specialized expertise in Electric Vehicles (EV), Hybrid Powertrains, and Model-Based Design (MBD) using MATLAB/Simulink.

Proven track record in field service, diagnostics, CAN tool analysis, and supporting product launches in both commercial and passenger EV segments.

Adept at root cause analysis, technical training, system simulations, and hands-on service operations.

PATENTS & PUBLICATIONS

Leakage Monitoring and Maintenance System in Water Lines

Patent Application No:
201941024633 A
Issue No: 26/2019

WORK EXPERIENCE

BYD India Pvt Ltd

Graduate Engineering Trainee, 2022 - 2023

Provided technical and service support for 200+ BYD electric trucks across Tamil Nadu, and Gujarat. Performed troubleshooting, preventive maintenance, technician training, dealer support, inventory management, and customer service, improving vehicle reliability, uptime, and operational efficiency while supporting continuous product improvement.

Electra EV

Technical Service Engineer, 2023 - 2026

Provided field service and remote diagnostics for electric vehicles, performed root-cause analysis, and escalated critical issues to engineering teams. Conducted technical training for dealers and ITI engineers, supported EV battery diagnostics using CAN tools, and assisted with pre-delivery inspections, new product launches, and technical documentation.

Octillion Power Systems

Engineer, 2026 - Present

Provided service support and diagnostics for lithium-ion EV battery systems, including battery pack troubleshooting, failure analysis, and root-cause investigations. Supported field teams on BMS and high-voltage system issues, collaborated with engineering and R&D teams, and analyzed service data to improve battery reliability, performance, and safety.

EDUCATION

Skill-Lync

Postgraduate Degree, 2022 - 2023

Hybrid Electrical Vehicle Design and Analysis and Completed many projects and Courses like MATLAB, Simulink, Development of Hybrid Electrical vehicle, Battery Technology for Electric Vehicle and Currently working on Model Based Development.

Misrimal Navajee Munoth Jain Engineering College

Bachelor of Engineering - BE, 2016 - 2020

Graduated with a Bachelor of Engineering in Mechanical Engineering, gaining expertise in automotive engineering, powertrain systems, materials, manufacturing, and maintenance practices. Developed strong analytical and troubleshooting skills through technical projects, which laid the foundation for a career in electric vehicle service, diagnostics, and battery technology.

SOFTWARE SKILLS

- MATLAB
- Simulink
- Ansys workbench
- CAD
- 3D CAD
- Openfoam
- Python
- P-CAN
- Busmaster
- Ms Excel
- Ms Word
- Teams

Dhilipan Sivasamy

ENGINEER



SUMMARY

A passionate and results-driven Mechanical Engineer with specialized expertise in Electric Vehicles foundation in MATLAB/Simulink, EV powertrain diagnostics, and field service engineering. Proven ability to troubleshoot complex issues, deliver technical training, and support product launches in both commercial and passenger EV segments.

TECHNICAL SKILLS

EV Systems & Diagnostics

- Model-Based Development (MBD)
- MIL, SIL & Code Generation
- Powertrain Architecture
- Battery Management Systems
- Autonomous Vehicle Systems
- Diagnostic Tool Support
- Root Cause Analysis using CAN-based Tools

Simulation & Programming

- MATLAB, Simulink, Simscape, Stateflow & Embedded C
- Thermal & Structural Analysis
- ANSYS CONVERGE CFD

Productivity & Office Tools

- MS Excel
- MS PowerPoint
- MS Word

PROJECTS

Vehicle Direction Determination

MATLAB | Simulink | MBD

Identifying the direction of the vehicle is one of the important & diverse features in Autonomous driving & Advanced Driver Assistance Features. This particular sub-feature of identifying the direction of vehicle is basically identifying the direction the vehicle is taking based on the camera input.

Adaptive Cruise Control

MATLAB | Simulink | Stateflow

Adaptive Cruise Control (ACC) automatically adjusts a vehicle's speed to match traffic flow while maintaining a safe distance from the vehicle ahead. By reducing the need for constant acceleration and braking in traffic, ACC enhances driving comfort, lowers driver stress, improves safety, and allows better focus on the road.

Electric Vehicle Powertrain Design

MATLAB | Simulink

Create a MATLAB model of electric car which uses a battery and a DC motor. Choose suitable blocks from Powertrain block set. Prepare a report about the Model. Making a Model of a EV with Dc motor And Battery which is a rear-wheel-Drive Vehicle.

Thermal modeling of Battery

MATLAB | Simulink

Thermal Modeling of Battery pack for a 10 cell series lithium ion battery model, simulate the thermal effects and compare life cycle performance at various temperatures, charge & discharge rates using MATLAB.

Mechanical Design of Battery

MATLAB | Simulink

Detail Preparation of Battery pack capacity: 18 kWh
Cell: ANR26650M1-B

Prepare a detailed battery pack drawing along with its enclosure are taken into

Design and Analysis of Propeller Shaft

College Project

Analyzed and replaced materials for improved corrosion resistance and mechanical performance.

BAJA ATV Model

AUTO CAD | Solidworks | 3D CAD

Designed and developed an All-Terrain Vehicle (ATV) powered by a Continuously Variable Transmission (CVT). The project focused on off-road vehicle dynamics, drivetrain design, transmission optimization, and performance analysis. Gained practical experience in vehicle design, fabrication, testing, and integration of mechanical systems for varying terrain conditions.